1 Estimating the SLR Parameters

Introduction

- OLS: choose β_0, β_1 to minimize $\sum_i (y_i - \beta_0 - \beta_1 x_i)^2$
- MLE: given a full distributional assumption on the errors, choose parameter values that make the observed data most probable

The Likelihood Function

The likelihood function is the joint density of the observed data, viewed as a function of the parameters rather than the data.

Since $Y_i \sim N(\beta_0 + \beta_1 x_i,\ \sigma^2)$, and due to the observations being independent, the likelihood function is:

$$L(\beta_0, \beta_1, \sigma^2;\ \mathbf{y}, \mathbf{x}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right)$$

Since ln is monotonically increasing, maximizing L is equivalent to maximizing the log-likelihood:

$$\ell(\beta_0, \beta_1, \sigma^2;\ \mathbf{y}, \mathbf{x}) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_i (y_i - \beta_0 - \beta_1 x_i)^2 \tag{1}$$

Maximizing Over β_0 and β_1

- For fixed σ^2, only the last term of ℓ in Equation (1) depends on β_0, β_1
- Maximizing ℓ over $\beta_0, \beta_1 \iff$ minimizing $\sum_i (y_i - \beta_0 - \beta_1 x_i)^2$, which is the same as the OLS criterion

  Question: what does this imply about $\hat{\beta}_{0,\mathrm{ML}}, \hat{\beta}_{1,\mathrm{ML}}$ versus the OLS estimates?

- They are identical — under the normal-error model, the the maximum likelihood estimates of β_0 and β_1 coincide exactly with the least-squares estimates

2 Estimating the Error Variance

MLE of σ^2

Differentiating ℓ (Equation (1)) with respect to σ^2 and solving:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\mathrm{SS}_{\mathrm{Res}}}{2\sigma^4} = 0$$
$$\hat{\sigma}^2_{\mathrm{ML}} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n} \tag{2}$$

- Compare to the OLS/unbiased estimator: $\mathrm{MS}_{\mathrm{Res}} = \mathrm{SS}_{\mathrm{Res}}/(n-2)$
- $\hat{\sigma}^2_{\mathrm{ML}}$ divides by n, not $n-2$ — it is *biased*: $\mathrm{E}[\hat{\sigma}^2_{\mathrm{ML}}] = \frac{n-2}{n}\sigma^2$
- $\mathrm{MS}_{\mathrm{Res}}$ is the bias-corrected version of $\hat{\sigma}^2_{\mathrm{ML}}$

The Difference, in Our Example

Rocket-propellant data: $n = 20$, $\mathrm{SS}_{\mathrm{Res}} = 166{,}254.65$

$$\hat{\sigma}^2_{\mathrm{ML}} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n} = \frac{166{,}254.65}{20} = 8312.73 \qquad \text{by Equation (2)}$$

$$\mathrm{MS}_{\mathrm{Res}} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n-2} = \frac{166{,}254.65}{18} = 9236.37$$

- The MLE underestimates σ^2 by about 10% here — $n = 20$ is small enough for the bias to matter
- The two converge as n grows

Where MLE and OLS Diverge, in General

	OLS	MLE
Needs a distribution?	No (mean, variance only)	Yes (full distribution)
β_0, β_1 estimates	Least squares	Identical, under normality
σ^2 estimate	Unbiased, $\mathrm{SS}_{\mathrm{Res}}/(n-2)$	Biased, $\mathrm{SS}_{\mathrm{Res}}/n$
Framework	Geometric	Probabilistic
Optimality	Gauss–Markov (BLUE)	Asymptotic efficiency

3 Comparison

Statistical Advantages of MLE

- Asymptotically efficient: attains the Cramér–Rao lower bound as $n \to \infty$, under correct specification
- Provides a general toolbox: standard errors from the Fisher information, likelihood-ratio/Wald/score tests, AIC/BIC for model comparison
- Generalizes to models for which OLS has no natural analogue, for example: logistic regression, Poisson regression, and other generalized linear models (GLM)
- Consistent and asymptotically normal under mild regularity conditions

Where OLS Still Wins

- Gauss–Markov: OLS is BLUE assuming only zero-mean, constant-variance, uncorrelated errors — *no* distributional assumption required
- MLE's efficiency and consistency rely on the assumed distribution being (at least approximately) correct
- Here, $\hat{\beta}_0, \hat{\beta}_1$ are unbiased in both approaches, but $\hat{\sigma}^2_{\mathrm{ML}}$ is biased while $\hat{\sigma}^2_{\mathrm{OLS}} = \mathrm{MS}_{\mathrm{Res}}$ is not